

CHETUMAL INTL, Mexico

WMO: 767500

Lat: 18.5006N

Lon: 88.3278W

Elev: 9

StdP: 101.22

Time Zone: -5.00 (W05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	15.4	17.3	12.9	9.3	18.1	14.4	10.2	19.1	10.4	26.2	10.2	26.6	1.8	330	0.370

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.6	34.3	26.9	33.7	26.9	33.0	26.7	28.2	32.3	27.7	31.9	27.5	31.6	3.5	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.1	22.9	31.2	26.7	22.3	30.8	26.2	21.6	30.2	91.1	32.4	88.6	32.1	87.1	31.9	32.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.3	10.0	9.5	12.1	37.3	1.2	1.1	11.2	38.1	10.5	38.7	9.8	39.3	9.0	40.1		
(5)			11.5	30.5	1.3	1.1	10.5	31.2	9.8	31.9	9.0	32.5	8.1	33.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	27.5	24.3	25.7	26.6	28.4	29.3	29.0	29.3	29.5	29.0	27.7	25.9	25.1		
	DBStd	2.50	2.20	2.20	2.14	1.57	1.31	1.44	1.42	1.42	1.38	1.67	1.70	2.09		
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDD10.0	6382	443	439	515	552	598	570	598	604	570	547	476	469		
	CDD18.3	3341	185	207	257	302	339	320	340	346	320	289	226	211		
	CDH23.3	33952	1240	1726	2361	3207	3924	3675	4018	4114	3592	2791	1784	1521		
	CDH26.7	12751	272	456	680	1140	1638	1493	1734	1858	1518	1026	533	402		
(14) Wind	WSAvg	3.6	3.2	3.8	4.3	4.8	4.6	4.1	3.7	3.6	3.1	2.8	2.6	2.7		
(15) Precipitation	PrecAvg	1306	67	30	35	37	106	199	125	163	178	195	105	66		
	PrecMax	1886	199	85	103	139	335	375	347	492	383	593	240	182		
	PrecMin	950	14	0	6	0	0	59	9	42	57	34	17	10		
	PrecStd	221	42	19	26	30	85	94	82	112	83	130	58	43		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	31.1	32.2	32.8	33.7	34.4	34.2	34.7	35.4	34.8	34.0	32.5	31.8		
		MCWB	25.0	24.8	25.5	26.3	26.8	27.0	26.9	26.9	27.2	26.9	25.8	25.6		
	2%	DB	30.1	31.1	31.8	32.7	33.5	33.2	33.8	34.2	33.9	32.9	31.4	30.8		
		MCWB	25.0	25.4	25.2	26.1	26.6	27.1	26.9	27.0	27.1	26.5	25.7	25.4		
	5%	DB	29.2	30.2	30.9	31.9	32.7	32.2	32.9	33.1	32.8	31.9	30.7	30.0		
		MCWB	24.6	25.1	25.2	25.9	26.5	26.9	26.7	27.0	27.0	26.5	25.6	25.1		
	10%	DB	28.2	29.2	30.0	31.0	31.8	31.5	32.0	32.2	31.9	31.0	29.9	29.0		
		MCWB	24.3	24.5	25.0	25.9	26.4	26.8	26.6	26.8	26.9	26.5	25.3	24.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	26.5	26.8	27.2	27.8	28.6	28.5	28.2	28.7	28.7	28.2	27.2	26.8		
		MCDB	29.5	30.5	31.2	31.9	32.6	32.1	32.7	33.0	32.8	31.9	31.0	30.4		
	2%	WB	25.7	26.2	26.4	27.1	27.7	27.7	27.6	27.9	28.0	27.5	26.7	26.2		
		MCDB	28.9	29.8	30.3	31.2	31.8	31.6	32.1	32.3	32.1	31.2	30.1	29.6		
	5%	WB	25.3	25.7	25.9	26.6	27.2	27.3	27.2	27.5	27.5	27.0	26.2	25.7		
		MCDB	28.4	29.1	29.6	30.5	31.2	31.1	31.5	31.8	31.5	30.7	29.5	28.9		
	10%	WB	24.8	25.2	25.4	26.2	26.7	27.0	26.9	27.1	27.1	26.6	25.7	25.2		
		MCDB	27.8	28.6	29.0	30.0	30.6	30.7	30.9	31.3	31.0	30.0	28.8	28.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.9	8.7	8.4	7.5	7.0	6.6	7.2	7.6	7.8	8.0	8.8	8.7		
		MCDBR	9.0	8.6	8.4	7.7	7.4	7.1	7.8	8.3	8.4	8.6	9.5	9.0		
	5% WB	MCWBR	5.4	4.8	4.5	3.9	3.5	3.2	3.3	3.7	3.7	4.2	5.1	5.1		
		MCWBR	8.2	8.1	7.7	7.2	7.0	6.8	7.4	8.0	7.9	8.1	8.5	8.4		
(40) Clear-Sky Solar Irradiance	taub	taub	0.369	0.381	0.412	0.459	0.483	0.467	0.495	0.468	0.439	0.411	0.389	0.382		
		taud	2.551	2.500	2.397	2.258	2.220	2.300	2.185	2.283	2.376	2.472	2.503	2.522		
	Ebn at Noon	Ebn at Noon	916	921	899	853	823	828	807	835	861	878	886	889		
		Edh at Noon	97	107	122	141	145	132	149	136	123	108	101	97		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	4.13	4.91	5.63	6.07	5.70	5.14	5.49	5.46	5.06	4.65	4.33	3.98		
	RadStd	RadStd	0.28	0.29	0.32	0.28	0.40	0.40	0.41	0.34	0.36	0.33	0.24	0.27		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page